

VIRGINIA MWEGA

Full-Stack and AI Engineer

virginiamwega2@gmail.com | +254 724 894 850 | <https://github.com/virginiamwega2-svg> | <https://linkedin.com/in/virginia-mwega> | Nairobi, Kenya

PROFILE SUMMARY

Full-Stack and AI Engineer with a track record of shipping production systems built on Claude tool use, structured outputs, response caching, and graceful fallbacks: AI that keeps working when an API key runs out of credits. Three production AI products delivered end-to-end (schema, API, AI integration, auth, payments, CI), serving real users with measurable outcomes. Focus areas: family wellness, hiring, and ecommerce.

EXPERIENCE

Solo Full-Stack and AI Engineer, Fit Parent Plan

2026 - Present

<https://fit-parent-plan-platform.vercel.app> | AI-powered fitness coaching platform for busy parents

- Owned the full product from rough idea to revenue: schema, AI agents, Stripe, custom JWT auth, CRO landing, event pipeline, and deploy.
- Shipped two production Claude agents (conversational 20-Minute Window workout planner and Pantry-to-Plate dinner agent) with streaming responses and mock fallback for zero-balance API resilience.
- Designed event pipeline emitting structured *logSession* events consumed by n8n automation that drafts and sends a weekly Parent Pulse digest.
- Engineered no-auth resilience: IP-based rate limiting with x-forwarded-for parsing, pagehide listener that auto-saves plans before browser close, and plan-pause via localStorage preserving 7-day breaks across sessions.
- Integrated Stripe Checkout with idempotent webhook handlers and a 3-question lead-qualification quiz filtering prospects before they reach the coach.
- Outcomes: 74 parents coached, 4.9 / 5 across 38 reviews, \$199/mo price point.

Stack: Next.js 15, TypeScript, Tailwind v4, Claude API, Stripe, JWT auth, SQLite, n8n, Playwright, Vercel.

Solo Builder, Hirely

2024 - 2025

<https://hirely-a0lx.onrender.com> | Full-stack AI hiring platform for parents and caregivers

- Designed and shipped 10+ production AI features end-to-end: conversational search grounded in live database via Claude tool use, CV parsing into a structured ParentProfile, AI-ranked top matches, employer auto-screening, and AI-drafted empathy emails on status change.
- Made job flexibility a first-class database enum (Fixed / Flexible / Async), powering reliable filtering, matching, and AI scoring across the platform.
- Engineered AI response caching by (user, profile.parsed_at) so re-uploaded resumes invalidate cleanly without time-based staleness.
- Built graceful fallbacks for every AI surface, keeping the platform functional when credits run out.
- Shipped weekly digest and stale-job nudge cron jobs, CI-gated deploys via GitHub Actions, and Neon Postgres on a zero-ops stack.

Stack: Django, PostgreSQL (Neon), Claude API, Render, GitHub Actions, Resend, WhiteNoise.

Solo Full-Stack Engineer, PureNest Family

2023

<https://purenest-family.vercel.app> | Family wellness ecommerce platform, built from scratch without a starter template

- Built a complete ecommerce platform end-to-end: schema, API, AI assistant, checkout, and admin tooling.
- Engineered idempotent Stripe webhook handling: processed event IDs persisted so retries never create duplicate orders, regardless of network failures.
- Integrated a Claude-powered conversational shopping assistant for contextual product recommendations across 5 wellness categories.
- Built full auth from scratch: email and password, OAuth, email verification, and signed one-time reset tokens.
- Shipped a server-validated coupon engine (percentage and fixed, usage limits, min subtotals, expiry) and verified-purchase reviews with database-level constraints on helpful votes.
- Hardened for production: locked-down security headers (CSP, HSTS, X-Frame-Options DENY) and admin dashboard with full audit log.

Stack: Next.js 16, React 19, TypeScript, PostgreSQL, Prisma, NextAuth, Stripe, Claude API, Resend, Tailwind v4, Vitest, Vercel.

SKILLS

Languages: TypeScript, Python, SQL, HTML, CSS

Frameworks: Next.js 15, React 19, Django, Tailwind CSS v4

AI Engineering: Claude API (tool use, streaming, structured outputs), response caching by user + parsed_at, graceful fallback patterns, prompt-led integration, conversational agents

Backend: PostgreSQL, Prisma, SQLite, NextAuth, custom JWT auth, REST API design, schema-first development

Payments and Automation: Stripe Checkout, idempotent webhook handling, processed-event-ID dedupe, n8n workflows, webhook pipelines, cron schedules

DevOps: Vercel, Render, GitHub Actions, CI-gated deploys, Playwright, Jest, Vitest

Production Hardening: rate limiting, security headers (CSP, HSTS, X-Frame-Options), email/OAuth/JWT auth flows, audit logs

EDUCATION

Self-Taught Software Engineering

2023 - Present

Independent study via shipping three production AI products end-to-end. See Experience and Certifications sections.

PUBLICATIONS

- Idempotent Stripe webhooks: the 30-line pattern that survives every retry. *Hashnode*, 2026.
- I asked 74 parents when they actually train. *Medium*, 2026.

CERTIFICATIONS

Full-Stack Engineering Certification

2024 - 2025

freeCodeCamp

Online